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NEPHROTIC AND NEPHRITIC SYNDROME

Nephrotic Syndrome

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Features :

1. massive proteinuria

(Hall mark)

- Excretion $\left\{ \begin{array}{l} > 40 \text{ mg/m}^2/\text{hr of urine excretion.} \\ > 1 \text{ g/m}^2/\text{day.} \end{array} \right.$

- Spot dipstick : 3+ or 4+.

- Protein : creatinine (P : C) ratio > 2 .

2. Hypoalbuminemia

Serum albumin $< 3 \text{ g/dl}$. $\downarrow$ (oncotic pressure)

3. Anasarca.

4. Hyperlipidemia (D/t $\uparrow$ production of lipoprotein and lipid by liver).

Pathology :

- minimal change disease (MCD) : m/c.

- Significant glomerular lesions :

- Focal segmental glomerulosclerosis (FSGS).

- mesangial proliferative glomerulonephropathy.

- C3 glomerulopathy.

CONDITIONS SEEN :

minimal change disease :

Pathogenesis :

Effacement of negatively charged foot process of podocyte

Negatively-charged protein not repelled

 $\uparrow$ excretion of proteins

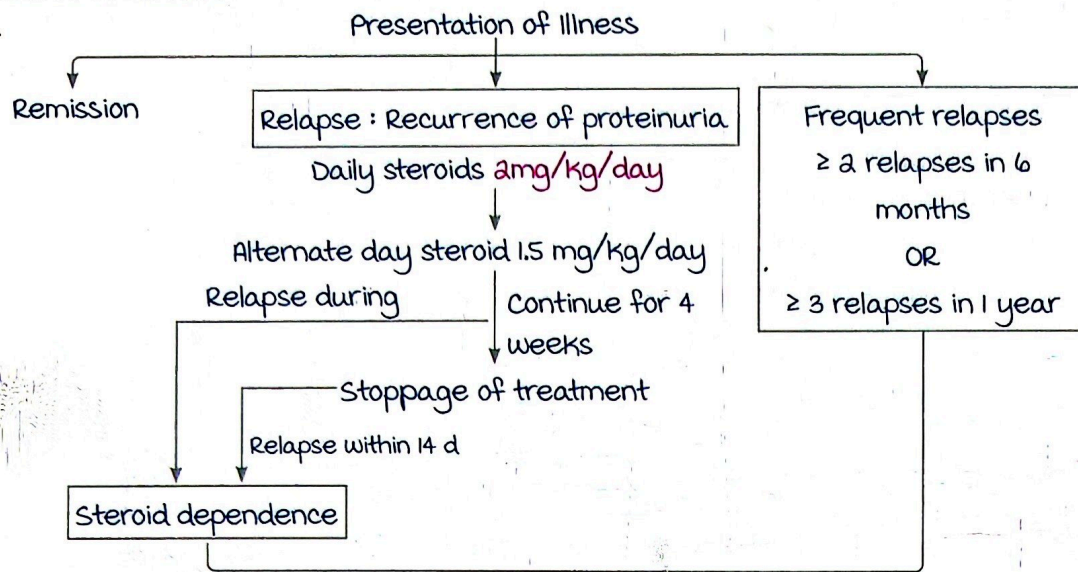
management :

Diet	- High protein diet. - No added salt.				
Steroid	- Prednisolone : <table border="1"> <tr> <td>6 weeks</td><td>6 weeks</td></tr> <tr> <td>2 mg/kg/day daily (upto 60 mg)</td><td>1.5 mg/kg/d on alternate days (upto 40 mg)</td></tr> </table> - Response (Confirms MCD without biopsy) : - Protein in urine $< 4 \text{ mg/m}^2/\text{hr}$ OR dipstick : Nil/trace. - P : C ratio < 0.2.	6 weeks	6 weeks	2 mg/kg/day daily (upto 60 mg)	1.5 mg/kg/d on alternate days (upto 40 mg)
6 weeks	6 weeks				
2 mg/kg/day daily (upto 60 mg)	1.5 mg/kg/d on alternate days (upto 40 mg)				
Diuretic (For significant edema)	Furosemide $\xrightarrow{\text{If no response}}$ Add Spironolactone				

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Course of disease :

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management

- Avoid steroids (Can lead to toxicity).
- Immunomodulatory drugs used :
 - Levamisole.
 - mycophenolate.

Steroid resistant nephrotic syndrome :

usually d/t significant glomerular lesions/congenital nephrotic syndrome (Onset >3 months).

Investigations :

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- Renal biopsy.
- Glomerular filtration rate (GFR) estimation.
- Genetic testing :

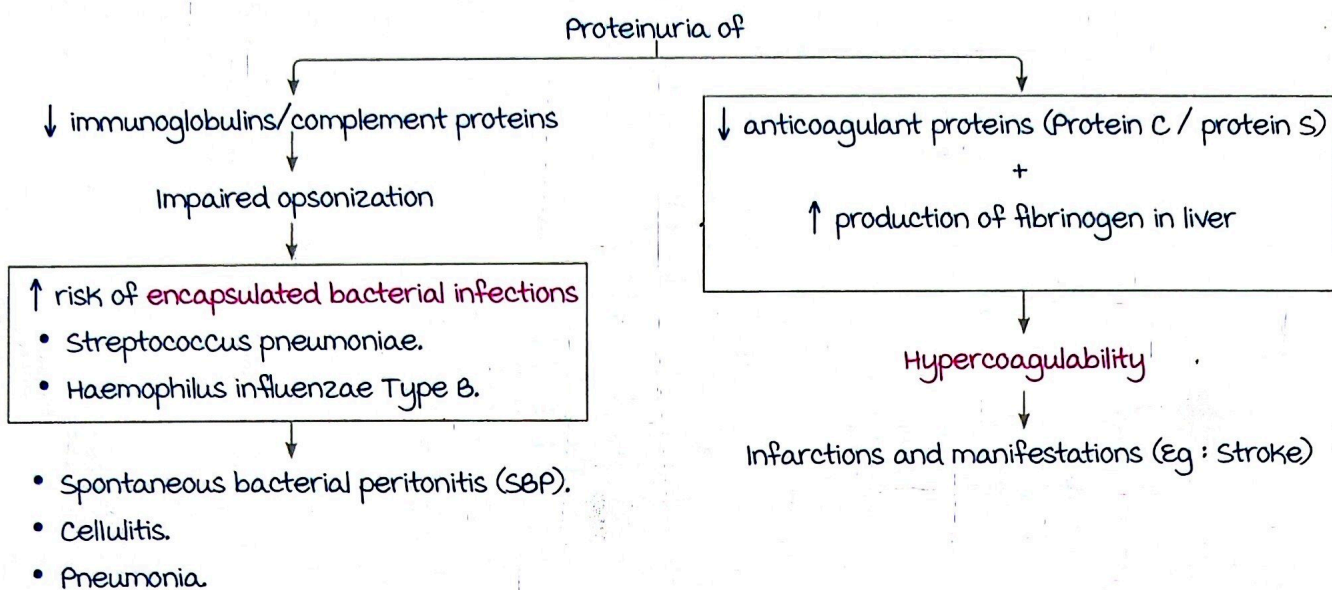
Common mutations in congenital nephrotic syndrome

NPHS1 gene (Finnish type).	NPHS2 gene	WT1 gene (Denys-Drash syndrome)
• Codes for nephrin. • Autosomal recessive.	• Codes for podocin. • Autosomal recessive.	• Autosomal dominant. • Associated with : - ↑ risk of wilms tumor. - Ambiguous genitalia in male.

management :

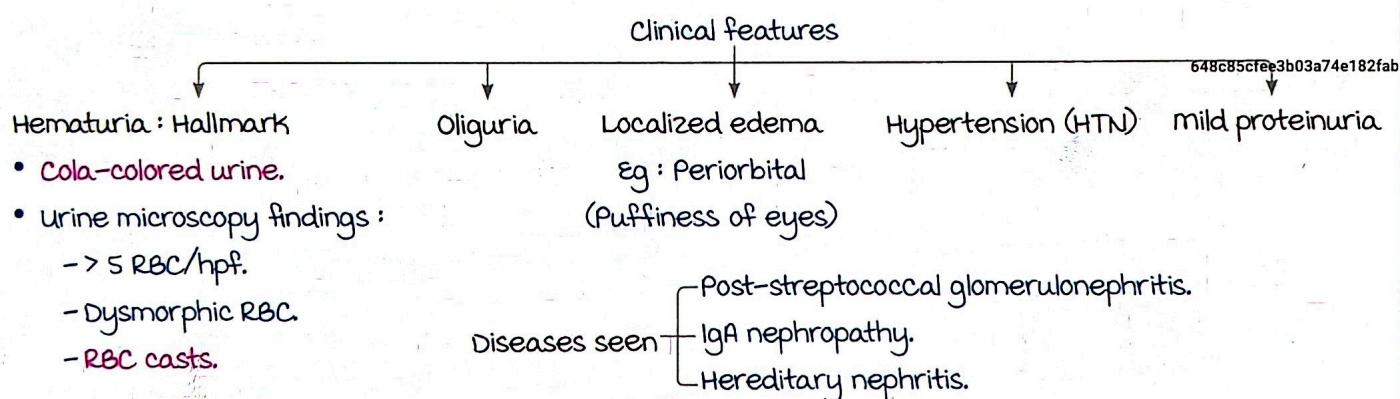
Calcineurin inhibitors (Immunomodulatory) : Cyclosporin, Tacrolimus.

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Nephritic Syndrome Glomerulonephritis

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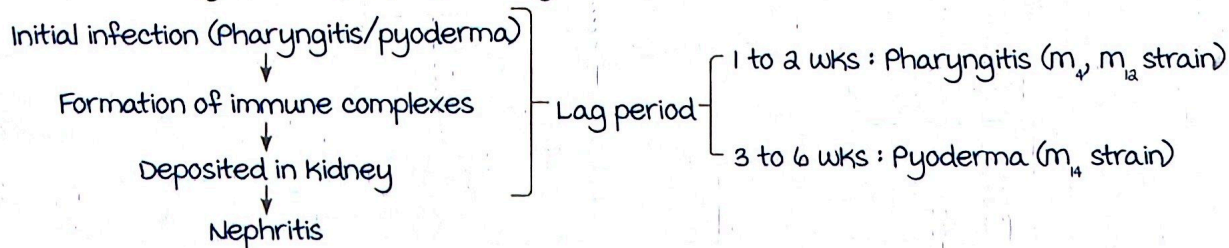


POST-STREPTOCOCCAL GLOMERULONEPHRITIS (PSGN) :

Post-infectious phenomenon :

Course : 7 - 10 days.

Pathogenesis : Group A β hemolytic streptococci.



Clinical features :

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- Incidence :
 - m/c age group : 5 - 15 years (School-going child).
 - male > female.
- Clinical presentation :
 - **Abrupt** onset of **hematuria, oliguria, edema** and **HTN**.
 - Preceding h/o infection not requisite for diagnosis.
 - Degree of oliguria decides disease severity.
- Complications :
 - Hypertensive encephalopathy : Seizures.
 - Left ventricular failure.

Investigations :

1. Antibody titre :

Estimation of :

- **Anti-Streptolysin O (ASO)**.
- Anti-DNAase B (Additionally in skin infection).

2. Serum C3 :

- ↓ in immune complex-mediated disorders.
- Value usually **normalises** within 8 - 12 weeks.

3. Renal biopsy : **Not performed routinely.**

• Indications :

- Persistent low C3 > 12 weeks (To rule out other causes).
- Systemic illness : High fever, rash, joint pain.
- No resolution > 10 days.

• Findings :

- Electron microscopy : Sub-epithelial immune complexes deposits → **Humps**.
- Immunofluorescence : IgG + C3 deposition in capillary walls and mesangium → **Starry sky appearance**.

management : Self-limiting condition

- Supportive management for HTN : Amlodipine, Nifedipine.
- **No prophylactic role** of antibiotics (Unlike rheumatic fever).

IgA NEPHROPATHY : (Berger's disease) .

- m/c cause of **chronic & recurrent nephritis** (Associated with CKD).

Pathogenesis :

Any viral/ bacterial upper respiratory tract infection (URTI)



IgA in mesangium & capillary walls

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Genitourinary System

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Investigations :

1. C3 levels : Normal.
2. ASO titre : No increase.
3. Renal biopsy : **mesangial proliferation + IgA deposits.**

management :

- Chronic nephropathy $\xrightarrow{10 - 20 \% \text{ cases}}$ Chronic Kidney disease
(Requires constant follow-up).
- ACE inhibitors/ ARB $\rightarrow$ $\downarrow$ renal progression of disease $\rightarrow$ $\downarrow$ risk of CKD.
- **Fish oil (Omega 3 PUFA)** $\rightarrow$ $\downarrow$ macrophage entry into mesangium $\rightarrow$ No initiating factors for IgA deposition.

HEREDITARY NEPHRITIS :

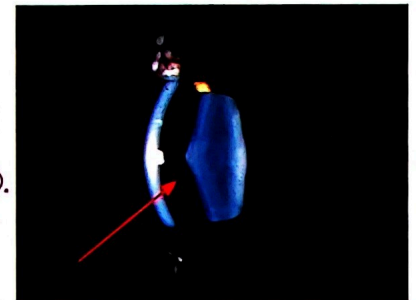
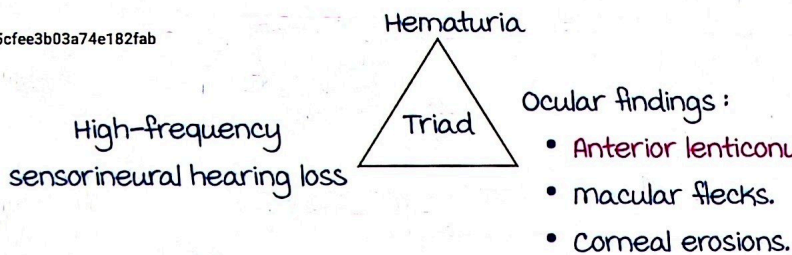
AKA Alports syndrome.

m/c mode of inheritance : X-linked dominant.

Genetic defect : **COL4A5 gene** ($\alpha 5$ chain of type IV collagen).

Clinical features :

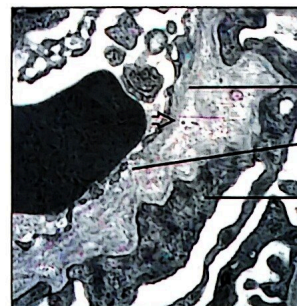
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Anterior lenticonus

Investigations :

1. Gene testing COL4A5 gene mutation.
2. Renal biopsy.



Renal biopsy

- Lamellations : Basket weave appearance
- Thinning of GBM
- Thickening of glomerular basement membrane (GBM)

management :

- Regular follow up : many develop end stage renal disease by 2nd - 3rd decade.
- Supportive treatment.
- ACE inhibitors $\rightarrow$ $\downarrow$ renal progression.